18/11/2022

UNIT-3

Complex Integration

Cauchy integral formula - Cauchy residue theorem.

contour integration

Cauchy integral formula:

$$f(a) = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-a)}\, dz$$

Problems:

1) Evaluate using cauchy integral theorem for

$\int_C \frac{f(z)}{z+4}$ where C is $|z| = 2$

To find singular points:

Given function fails to be analytic

∴ denomination is equal to zero.

$$z+4=0$$

$$\boxed{z=-4}$$

Gn:

$$|z| = 2$$

$$|-4| = 2$$

$$4 > 2 \notin C$$

$$\therefore \boxed{\int_C \frac{f(z)}{z+4} = 0}$$

|z| = 2
z value is less than 2 then proceed
if greater than 2 put 0 as answer

2) Evaluate $\int_c \frac{3z-1}{z^3-z}\,dz$ where, c is $|z|=2$

Soln:

To find singular points:

Given function fails to be analytic therefore denominator is equal to zero.

$$z^3 - z = 0$$

$$z(z^2-1) = 0$$

$$z = 0 \quad \& \quad z^2 = 1$$

$$z = \pm\sqrt{1}$$

$$z = \pm 1$$

$$\therefore \boxed{z = 0, 1, -1}$$

Gn:

$$|z| = 2$$

$\boxed{z=0}$

$$|0| = 2$$
$$0 < 2 \in C$$

$\boxed{z=-1}$

$$|-1| = 2$$
$$1 < 2 \in C$$

$\boxed{z=1}$

$$|1| = 2$$
$$1 < 2 \in C$$

we have to separate the denominator to solve the problem

$$\int \frac{3z-1}{z^3-z}\,dz = \int \frac{3z-1}{(z)(z-1)(z+1)}\,dz$$

$$= \underbrace{\int \frac{\frac{3z-1}{(z-1)(z+1)}}{z}\,dz}_{I_1} + \underbrace{\int \frac{\frac{3z-1}{(z)(z+1)}}{(z-1)}\,dz}_{I_2} +$$

$$\underbrace{\int \frac{\frac{3z-1}{(z)(z-1)}}{(z+1)}\,dz}_{I_3}$$

To find I_1 :

$$\int_c \frac{\frac{3z-1}{(z-1)(z+1)}}{z} \, dz = I_1$$

W.K.T

By C.I.F

$$\int_c \frac{f(z)}{(z-a)} \, dz = 2\pi i f(a)$$

$$f(z) = \frac{3z-1}{(z-1)(z+1)}$$

$$(z-a) = z$$

$$z - a - z = 0$$

$$\boxed{a = 0}$$

put value of a in f(z)

$$f(a) = \frac{3(0)-1}{(0-1)(0+1)} = \frac{-1}{(-1)(1)} = \frac{-1}{-1} = 1$$

$$I_1 = 2\pi i(1)$$

$$\boxed{I_1 = 2\pi i}$$

To find I_2 :

$$I_2 = \int_c \frac{\frac{3z-1}{(z)(z+1)}}{(z-1)} \, dz$$

W.K.T By C.I.F

$$\int_c \frac{f(z)}{(z-a)} \, dz = 2\pi i f(a)$$

$$f(z) = \frac{3z-1}{(z)(z+1)}$$

$$(z-a) = (z-1)$$

$$\boxed{a=1}$$

$$f(a) = \frac{3(1)-1}{(1)(1+1)} = \frac{2}{2} = 1$$

$$I_2 = 2\pi i(1)$$

$$\boxed{I_2 = 2\pi i}$$

To find I_3:

$$I_3 = \int_c \frac{\frac{3z-1}{(z)(z-1)}}{(z+1)} \, dz$$

W.K.T By C.I.F

$$\int_c \frac{f(z)}{(z-a)} dz = 2\pi i f(a)$$

$$f(z) = \frac{3z-1}{(z)(z-1)}$$

$$z-a = z+1$$

$$-a = 1$$

$$a = -1$$

$$f(a) = \frac{3(-1)-1}{(-1)(-1-1)} = \frac{-4}{2} = -2$$

$$I_3 = 2\pi i(-2)$$

$$\boxed{I_3 = -4\pi i}$$

$$\int_c \frac{3z-1}{z^3-z}\,dz = 2\pi i + 2\pi i - 4\pi i$$
$$= 4\pi i - 4\pi i$$

$$\boxed{\int_c \frac{3z-1}{z^3-z}\,dz = 0}$$

3) Using cauchy integral formula Evaluate

$$\int_c \frac{\sin\pi z^2 + \cos\pi z^2}{(z-1)(z-2)} \text{ where } c \text{ is } |z| = 3$$

To find singular point:

Given function fails to be analytic

$\therefore$ Denominator equal to 0

$$z-1=0 \quad\Big|\quad z-2=0$$
$$z=1 \qquad\qquad z=2$$

$$\therefore z = 1, 2$$

Gn:

$$|z| = 3$$

$[\because \sin\pi, 2\pi, 3\pi, 4\pi = 0$

odd $\rightarrow \cos\pi, 3\pi, 5\pi, \ldots = -1$

even $\rightarrow \cos 2\pi, 4\pi, 6\pi, \ldots = 1]$

$\boxed{z=1}$

$$|1| = 3$$
$$1 < 3 \in c$$

$\boxed{z=2}$

$$|2| = 3$$
$$2 < 3 \in c$$

$$\int_c \frac{\sin\pi z^2 + \cos\pi z^2}{(z-1)(z-2)}\,dz = \underbrace{\int_c \frac{\frac{\sin\pi z^2 + \cos\pi z^2}{(z-1)}}{(z-2)}\,dz}_{I_1} + \underbrace{\int_c \frac{\frac{\sin\pi z^2 + \cos\pi z^2}{(z-2)}}{(z-1)}\,dz}_{I_2}$$

To find I_1 :

$$I_1 = \int_C \frac{\frac{\sin\pi z^2 + \cos\pi z^2}{(z-1)}}{(z-2)} \, dz$$

W.K.T

By C.I.F

$$\int_C \frac{f(z)}{(z-a)} \, dz = 2\pi i f(a)$$

$$f(z) = \frac{\sin\pi z^2 + \cos\pi z^2}{(z-1)}$$

$$(z-a) = z-2$$

$$\boxed{a=2}$$

$$f(a) = \frac{\sin\pi(2)^2 + \cos\pi(2)^2}{(2-1)} = \frac{\sin 4\pi + \cos 4\pi}{1}$$

$$\cancel{= \frac{0-4}{1} = -4} \qquad = \frac{0+1}{1} = 1$$

$$I_1 = 2\pi i\,(\cancel{-4})$$

$$I_1 = \cancel{-8\pi i}\ 2\pi i$$

To find I_2 :

$$I_2 = \int_C \frac{\frac{\sin\pi z^2 + \cos\pi z^2}{(z-2)}}{(z-1)} \, dz$$

W.K.T

by C.F.T

$$\int_C \frac{f(z)}{(z-a)} \, dz = 2\pi i f(a)$$

$$f(z) = \frac{\sin\pi z^2 + \cos\pi z^2}{(z-2)}$$

$z - a = z - 1$

$\boxed{a = 1}$

$$f(a) = \frac{\sin\pi(1)^2 + \cos\pi(1)^2}{(1-2)}$$

$$= \frac{0 - 1}{-1} = 1$$

$I_2 = 2\pi i(1)$

$I_2 = 2\pi i$

$$\int_C \frac{\sin\pi z^2 + \cos\pi z^2}{(z-1)(z-2)} dz = I_1 + I_2$$

$= 2\pi i + 2\pi i$

$= 4\pi i$

4) using cauchy integral formula Evaluate $\int_C \frac{z+4}{z^2+2z+5} dz$, (A) 11m

c is $|z+1-i| = 2$

Soln:

To find Singular point :

Given function fails to be analytic

$\therefore$ denominator equal to 0

$z^2 + 2z + 5 = 0$

$a = 1, \; b = 2, \; c = 5$

$$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{2^2-4(1)(5)}}{2}$$

$$= \frac{-2 \pm \sqrt{4-20}}{2}$$

$$= \frac{-2 \pm \sqrt{-16}}{2}$$

$$= \frac{-2 \pm 4i}{2}$$

$$= -1 \pm 2i$$

$\therefore \; z = -1+2i, \; -1-2i$

Gn :

$$|z+1-i| = 2$$

put $z = -1+2i$

$$|\cancel{-1}+2i+\cancel{1}-i| = 2$$

$$|i| = 2$$

$$\sqrt{0^2+1^2}$$

$$1 < 2 \in C$$

Put $z = -1-2i$

$$|-\cancel{1}-2i+\cancel{1}-i| = 2$$

$$|-3i| = 2$$

$$\sqrt{0^2+(-3)^2}$$

$$\sqrt{9}$$

$$3 > 2 \notin C$$

$$\int_C \frac{z+4}{z^2+2z+5}\,dz = \int_C \frac{z+4}{(z+1-2i)(z+1+2i)}\,dz$$

$$= \underbrace{\int_C \frac{\frac{z+4}{(z+1+2i)}}{(z+1-2i)}\,dz}_{I_1} + \underbrace{\int_C \frac{\frac{z+4}{(z+1-2i)}}{(z+1+2i)}\,dz}_{I_2}$$

To find I_1 :

$$\int_C \frac{\frac{z+4}{(z+1+2i)}}{(z+1-2i)}\,dz = I_1$$

W.K.T

C.I.F

$$\int_C \frac{f(z)}{(z-a)}\,dz = 2\pi i f(a)$$

$$f(z) = \frac{z+4}{(z+1+2i)}$$

$$(\cancel{z}-a) = \cancel{z}+1-2i$$

$$a = -1+2i$$

$$f(a) = \frac{(-1+2i)+4}{(-1+2i+1+2i)} = \frac{3+2i}{4i} = \cancel{-\frac{3i}{4}} + \cancel{\frac{2i}{4i}}$$

$$= \cancel{-\frac{3i}{4}} + \cancel{\frac{1}{2}}$$

$$I_1 = \cancel{2}\pi \cancel{i}\left(\frac{3+2i}{\cancel{4}\cancel{i}_{2}}\right)$$

$$\therefore I_1 = \frac{(3+2i)\pi}{2}$$

To find I_2:

$$\int_c \frac{\frac{z+4}{(z+1-2i)}}{(z+1+2i)} \, dz = I_2$$

$z-a = z+1+2i$

$a = -1-2i \not\in c$

$\therefore I_2 = 0$

$$\int_c \frac{z+4}{z^2+2z+5} \, dz = I_1 + I_2 = \frac{(3+2i)\pi}{2} + 0$$

$$\boxed{\therefore \int_c \frac{z+4}{z^2+2z+5} \, dz = \frac{(3+2i)\pi}{2}}$$

5) Using cauchy intergal formula Evaluate $\int_c \frac{z+1}{z^2+2z+4} \, dz$

where, c is $|z+1+i| = 2$

soln:

To find singular point:

Given function fails to be analytic

$\therefore$ Denominator equal to zero.

$$z^2+2z+4 = 0$$

$a = 1, \quad b = 2, \quad c = 4$

$$z = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-2 \pm \sqrt{2^2-4(1)(4)}}{2} = \frac{-2 \pm \sqrt{4-16}}{2}$$

$$= \frac{-2 \pm \sqrt{-12}}{2}$$

$$= \frac{-2 \pm i2\sqrt{3}}{2}$$

$$z = -1 \pm i\sqrt{3}$$

$$\therefore \; z = -1 + i\sqrt{3} \;,\; -1 - i\sqrt{3}$$

Gn:

$$|z + 1 + i| = 2$$

$$\boxed{z = -1 + i\sqrt{3}}$$

$$|\cancel{-1} + i\sqrt{3} + \cancel{1} + i| = 2$$

$$|i(\sqrt{3}+1)| = 2$$

$$|i(1.73+1)| = 2$$

$$\sqrt{0^2 + (2.73)^2} = 2$$

$$2.73 > 2 \notin C$$

Put $z = -1 - i\sqrt{3}$

$$|-\cancel{1} - i\sqrt{3} + \cancel{1} + i| = 2$$

$$|i(1 - \sqrt{3})| = 2$$

$$|i(1 - 1.73)| = 2$$

$$\cancel{\sqrt{0^2 + (0.73)^2} = 2}$$

$$\cancel{0.27 < 2 \notin C}$$

$$\sqrt{0^2 + (0.73)^2} = 2$$

$$0.73 < 2 \in C$$

$$\begin{array}{r} 1.00 \\ 1.73 \\ \hline 0.73 \end{array}$$

$$\int_C \frac{z+1}{z^2+2z+4}\,dz = \int_C \frac{z+1}{(z+1-i\sqrt{3})(z+1+i\sqrt{3})}\,dz$$

$$= \underbrace{\int_C \frac{\frac{z+1}{(z+1-i\sqrt{3})}}{(z+1+i\sqrt{3})}\,dz}_{I_1} + \underbrace{\int_C \frac{\frac{z+1}{(z+1+i\sqrt{3})}}{(z+1-i\sqrt{3})}\,dz}_{I_2}$$

To find I_1 :

$$I_1 = \int_C \frac{\frac{z+1}{(z+1-i\sqrt{3})}}{(z+1+i\sqrt{3})}\,dz$$

W.K.T

C.I.F

$$\int_C \frac{f(z)}{(z-a)}\,dz = 2\pi i f(a)$$

$$f(z) = \frac{z+1}{(z+1-i\sqrt{3})}$$

$$\cancel{z}-a = \cancel{z}+1+i\sqrt{3}$$

$$a = -1-i\sqrt{3}$$

$$f(a) = \frac{\cancel{-1}-i\sqrt{3}+\cancel{1}}{\cancel{-1}-i\sqrt{3}+\cancel{1}-i\sqrt{3}}$$

$$= \frac{\cancel{-}\,i\cancel{\sqrt{3}}}{\cancel{-}\,2i\cancel{\sqrt{3}}} = \frac{1}{2}$$

$$I_1 = 2\pi i\left(\frac{1}{2}\right)$$

$$\boxed{I_1 = \pi i}$$

To find I_2 :

$$I_2 = \int_C \frac{\frac{z+1}{(z+1+i\sqrt{3})}}{(z+1-i\sqrt{3})} \, dz$$

W.K.T

C.I.F

$$\int_C \frac{f(z)}{(z-a)} \, dz = 2\pi i f(a)$$

$$f(z) = \frac{z+1}{z+1+i\sqrt{3}}$$

$$(\cancel{z}-a) = \cancel{z}+1-i\sqrt{3}$$

$$a = -1+i\sqrt{3} \Rightarrow \notin C$$

$$\therefore I_2 = 0$$

$$\int_C \frac{z+1}{z^2+2z+4} \, dz = I_1 + I_2$$

$$= \pi i + 0$$

$$\therefore \boxed{\int_C \frac{z+1}{z^2+2z+4} = \pi i}$$

6) Evaluate $\int_C \frac{z}{(z-1)(z-2)^2} dz$ where, c is $|z-2| = \frac{1}{2}$ using Cauchy Integral formula

Soln:

To find singular point:

Given function fails to be analytic

$\therefore$ denominator equal to zero.

Gn: $(z-1) = 0$

$z = 1$

$z - 2 = 0$

$z = 2$ (twice)

Gn:

$|z-2| = \frac{1}{2}$

put $z=1$ $\quad |1-2| = \frac{1}{2}$

$1 > \frac{1}{2} \notin C$

put $z=2$

$|2-2| = \frac{1}{2}$

$0 < \frac{1}{2} \notin C$

$$\int_C \frac{z}{(z-1)(z-2)^2} dz = \underbrace{\int_C \frac{\frac{z}{(z-1)}}{(z-2)^2} dz}_{I_1} + \underbrace{\int_C \frac{\frac{z}{(z-2)^2}}{(z-1)} dz}_{I_2}$$

To find I_1:

$$I_1 = \int_C \frac{\frac{z}{(z-1)}}{(z-2)^2} dz$$

W.K.T

CIF

$$\int_c \frac{f(z)}{(z-a)^2} = 2\pi i f'(a)$$ → only for $(\ )^2$

$$f(z) = \frac{z}{z-1}$$

$$(z-a)^2 = (z-2)^2$$

$$\boxed{a=2}$$

$$f'(z) = \frac{(z-1)(1) - z(1)}{(z-1)^2}$$

$$= \frac{\cancel{z}-1-\cancel{z}}{(z-1)^2}$$

$$= \frac{-1}{(z-1)^2}$$

$$f'(a) = \frac{-1}{(2-1)^2} = \frac{-1}{1} = -1$$

$$I_1 = 2\pi i(-1)$$

$$\therefore \boxed{I_1 = -2\pi i}$$

To find I_2:

$$I_2 = \int_c \frac{\frac{z}{(z-2)^2}}{z-1}\, dz$$

$f(z)$

W.K.T

CIF

$$\int_c \frac{f(z)}{(z-a)} dz = 2\pi i f(a)$$

$$f(z) = \frac{z}{(z-2)^2}$$

$$z-a = z-1$$

$$a = 1 \Rightarrow \notin c$$

$$\therefore \boxed{I_2 = 0}$$

$$\int_c \frac{z}{(z-1)(z-2)^2} dz = I_1 + I_2$$

$$= -2\pi i + 0$$

$$\therefore \boxed{\int_c \frac{z}{(z-1)(z-2)^2} = -2\pi i}$$

Q.P. Questions :-

1) If $f(a) = \int_c \frac{3z^2+7z+1}{z-a}$ where, c is circle of $|z|=2$

i) Find $f(3)$

ii) Find $f'(1-i)$

Soln:

i) To find $f(3)$:

$$f(3) = \int_c \frac{3z^2+7z+1}{z-3}$$

To find singular point:

Given function fails to be analytic

∴ Denominator equal to zero.

$$z-3=0$$

$$\boxed{z=3}$$

Gn:

$$|z|=2$$

$$|3|=2$$

$$3>2 \notin c$$

$$\therefore \boxed{f(3)=0}$$

ii) To find $f'(1-i)$

$$f'(a) = -\int_c \frac{3z^2+7z+1}{(z-a)^2}(-1)$$

$$f'(a) = \int_c \frac{3z^2+7z+1}{(z-a)^2}$$

$\left[\because d\left(\frac{1}{x}\right) = d(x^{-1}) = -\frac{1}{x^2}\right]$

$$f'(1-i) = \int_c \frac{3z^2+7z+1}{(z-1+i)^2}$$

where, $a = 1-i$

Gn:

$$|z|=2$$

$$|+1-i|=2$$

$$\sqrt{1^2+1^2}=2$$

$$\sqrt{2}=2$$

$$1.4<2 \in c$$

$$\therefore \int_C \frac{f(z)}{(z-a)^2} = 2\pi i f'(a)$$

$f(z) = 3z^2 + 7z + 1$

$f'(z) = 6z + 7$

$a = -i + 1$

$f'(a) = 6(1-i) + 7$

$= 6 - 6i + 7$

$= 13 - 6i$

$\Rightarrow 2\pi i \, f'(a)$

$\Rightarrow 2\pi i \, (13 - 6i)$

$\therefore \boxed{f'(1-i) = 2\pi i(13-6i)}$

Cauchy residue theorem :-

Formulae :-

$$\int_C f(z)dz = 2\pi i \,[\text{sum of residues}]$$

To calculate residues :

i) If $z = a$ is a simple pole (or) pole of order 1

$$[\text{Res}\, f(z)]_{z=a} = \lim_{z \to a} (z-a) f(z)$$

ii) $z = a$ of order m

$$[\text{Res}\, f(z)]_{z=a} = \frac{1}{(m-1)!} \lim_{z \to a} \left[\frac{d^{m-1}}{dz^{m-1}} (z-a)^m f(z) \right]$$

1) Find the residue of $f(z) = \frac{z^2}{(z+2)(z-1)^2}$ at all poles

Soln:

$z+2=0$

$\boxed{z=-2}$ → It is a pole of order 1

$z-1=0$

$\boxed{z=1}$ → It is a pole of order 2

$\boxed{z=-2}$ → order 1

$$[\text{Res } f(z)]_{z=-2} = \lim_{z \to -2} \cancel{(z+2)} \times \frac{z^2}{\cancel{(z+2)}(z-1)^2}$$

$$= \lim_{z \to -2} \frac{z^2}{(z-1)^2}$$

$$= \frac{(-2)^2}{(-2-1)^2}$$

$$\boxed{[\text{Res } f(z)]_{z=-2} = \frac{+4}{9}}$$

To find residue 2:

$z=1$ of order 2

$$[\text{Res } f(z)]_{z=1} = \frac{1}{(2-1)!} \lim_{z \to 1} \left[\frac{d}{dz} \cancel{(z-1)^2} \times \frac{z^2}{(z+2)\cancel{(z-1)^2}} \right]$$

$$= 1 \cdot \lim_{z \to 1} \left[\frac{d}{dz} \frac{z^2}{z+2} \right]$$

$$= \lim_{z \to 1} \left[\frac{(z+2)(2z) - z^2(1)}{(z+2)^2} \right]$$

$$= \lim_{z \to 1} \left[\frac{2z^2 + 4z - z^2}{(z+2)^2} \right]$$

$$= \lim_{z \to 1} \left[\frac{z^2 + 4z}{(z+2)^2} \right]$$

$$= \frac{1+4}{(3)^2} = \frac{5}{9}$$

$$\boxed{\left[\operatorname{Res}_{z=1} f(z) \right] = 5/9}$$

2) Evaluate $\int_C \frac{4-3z}{(z)(z-1)(z-2)}$ where C is $|z| = 3/2$ using cauchy residue theorem

Soln:

To find singular point:

Given function fails to be analytic $\therefore$ denominator equal to zero

$\boxed{z=0}$ of order 1

$z - 1 = 0$

$\boxed{z = 1}$ of order 1

$z - 2 = 0$

$\boxed{z = 2}$ of order 1

Gn

$|z| = \frac{3}{2}$

$\boxed{z=0}$

$|0| = \frac{3}{2}$

$0 < 3/2 \in C$

$\boxed{z=1}$ $|1| = 3/2$

$1 < 3/2 \in C$

$\boxed{z=2}$

$|2| = 3/2$

$2 > 3/2 \notin C$

To find residue :

$\underline{R_1}$ given $z = 2$

$2 \notin C$ (outside)

$\therefore \boxed{R_1 = 0}$

$\underline{R_2}$ $z = 0$

$$[\text{Res } f(z)]_{z=0} = \lim_{z \to 0} (z) \times \left(\frac{4 - 3z}{(z)(z-1)(z-2)} \right)$$

$$= \frac{4 - 0}{(-1)(-2)}$$

$$= \frac{4}{2}$$

$$\therefore \boxed{[\text{Res } f(z)]_{z=0} = 2}$$

$z = 1$

$$\left[\operatorname{Res} f(z)\right]_{z=1} = \lim_{z \to 1} \cancel{(z-1)} \times \frac{4-3z}{z \times \cancel{(z-1)}(z-2)}$$

$$= \frac{4-3}{(1)\cancel{(1-1)}(1-2)}$$

$$= \frac{1}{1 \cancel{(0)} (-1)}$$

$$= -1$$

$$\boxed{\left[\operatorname{Res} f(z)\right]_{z=1} = -1}$$

$$\therefore \int_C \frac{4-3z}{(z)(z-1)(z-2)} = 2\pi i \,(\text{Sum of Residue})$$

$$= 2\pi i (R_1 + R_2 + R_3)$$

$$= 2\pi i (2-1)$$

$$= 2\pi i$$

3) Using cauchy residue theorem Evaluate

$$\int_C \frac{\cos\pi z^2 + \sin\pi z^2}{(z+1)(z+2)} dz$$ where, C is $|z| = 3$

Soln:

To find singular point:

Functions fails to be analytic

$\therefore$ denominator equal to zero.

$z+1=0$

$\boxed{z=-1}$ of order 1

$z+2=0$

$\boxed{z=-2}$ of order 1

Gn:

$|z| = 3$

$\boxed{z=-1}$

$|-1| = 3$

$1 < 3 \in C$ (Inside)

$\boxed{z=-2}$ $|-2|=3$

$2<3 \in C$ (inside)

To find Residue:

<u>R1</u> $z=-1$

$$\left[\text{Res } f(z)\right]_{z=-1} = \lim_{z\to -1} \cancel{(z+1)} \times \frac{\cos\pi z^2 + \sin\pi z^2}{\cancel{(z+1)}(z+2)}$$

$$= \frac{\cos\pi(-1)^2 + \sin\pi(-1)^2}{(-1+2)}$$

$$= \frac{\cos\pi + \sin\pi}{1} = \frac{-1+0}{1} = -1$$

$$\boxed{\left[\text{Res } f(z)\right]_{z=-1} = -1}$$

<u>R2</u>

$z=-2$

$$\left[\text{Res } f(z)\right]_{z=-2} = \lim_{z\to -2} \cancel{(z+2)} \times \frac{\cos\pi z^2 + \sin\pi z^2}{(z+1)\cancel{(z+2)}}$$

$$= \frac{\cos\pi(-2)^2 + \sin\pi(-2)^2}{(-2+1)}$$

$$= \frac{\cos 4\pi + \sin 4\pi}{-1} = \frac{1+0}{-1} = -1$$

$$\boxed{\left[\text{Res } f(z)\right]_{z=-2} = -1}$$

$$\therefore \int_C \frac{\cos\pi z^2 + \sin\pi z^2}{(z+1)(z+2)} = 2\pi i \text{ [sum of residue]}$$

$$= 2\pi i [R_1 + R_2]$$

$$= 2\pi i [-1-1]$$

$$= 2\pi i(-2)$$

$$= -4\pi i$$

4) Using Cauchy residue theorem Evaluate $\int_C \frac{z+1}{z^2+2z+4} dz$

where, C is $|z+1+i| = 2$

Soln:

To find singular point:

Given function fails to be analytic

$\therefore$ Denominator equal to 0.

$$z^2+2z+4 = 0$$

$$\boxed{z = -1+i\sqrt{3}, \ -1-i\sqrt{3}}$$

Gn:

$$|z+1+i| = 2$$

$z = -1+i\sqrt{3}$ $\quad |\cancel{-1} + i\sqrt{3} + \cancel{1} + i| = 2$

$$\sqrt{0^2 + (2.73)^2} = 2$$

~~2.73>2~~

$$2.73 > 2 \notin C$$

$z = -1-i\sqrt{3}$

$$|z+1+i| = 2$$

$$|\cancel{-1} - i\sqrt{3} + \cancel{1} + i| = 2$$

$$|i(1-\sqrt{3})| = 2$$

$|i(-0.73)| = 2$

$\sqrt{0^2+(0.73)^2} = 2$

$0.73 < 2 \in C$

$$f(z) = \int_C \frac{z+1}{(z+1-i\sqrt{3})(z+1+i\sqrt{3})}$$

To find residue:

<u>R_1</u>

$z = -1+i\sqrt{3}$ is a pole of order 1 $\notin C$

$\therefore \boxed{R_1 = 0}$

<u>R_2</u>

$z = -1-i\sqrt{3}$ of order 1

$$[\text{Res } f(z)]_{z=-1-i\sqrt{3}} = \lim_{z\to(-1-i\sqrt{3})} (\cancel{z+1+i\sqrt{3}}) \times \frac{z+1}{(z+1-i\sqrt{3})(\cancel{z+1+i\sqrt{3}})}$$

$$= \frac{\cancel{-1}-i\sqrt{3}+\cancel{1}}{\cancel{-1}-i\sqrt{3}+\cancel{1}-i\sqrt{3}}$$

$$= \frac{\cancel{-i\sqrt{3}}}{\cancel{-2i\sqrt{3}}}$$

$\therefore R_2 = \frac{1}{2}$

$$\therefore \int_C \frac{z+1}{z^2+2z+4}\,dz = 2\pi i\,[\text{sum of residue}]$$

$= 2\pi i\,[R_1+R_2]$

$= 2\pi i\,[0+\frac{1}{2}]$

$= \cancel{2}\pi i\,(1/\cancel{2})$

$= \pi i$

5) Evaluate cauchy residue theorem for $\int_C \frac{dz}{(z^2+4)^2} dz$ where C is $|z-i|=2$

soln:

To find singular point:

Given function fails to be analytic

∴ denominator equal to zero.

$$z^2+4=0$$
$$z^2=-4$$
$$z=\pm\sqrt{-4}$$
$$z=\pm 2i$$ it is a pole of order 2

$$\therefore z=-2i, +2i$$

Gn:

$$|z-i|=2$$

put $z=-2i$

$$|-2i-i|=2$$
$$|-3i|=2$$
$$\sqrt{0^2+3^2}=2$$
$$\sqrt{9}=2$$
$$3>2 \notin C$$

Put $z=2i$

$$|2i-i|=2$$
$$|i|=2$$
$$\sqrt{0^2+1^2}=2$$
$$1<2 \in C$$

$$f(z) = \int_C \frac{dz}{(z-2i)^2(z+2i)^2}$$

To find residue:

<u>R1</u>

$z = -2i$ of order '2'

$z = -2i \notin C$

$\therefore \boxed{R_1 = 0}$

<u>R2</u>

$z = 2i$ of order '2'

$$[\text{Res}\, f(z)]_{z=2i} = \frac{1}{1!} \lim_{z \to 2i} \left[\frac{d}{dz} \cancel{(z-2i)^2} \times \frac{1}{\cancel{(z-2i)^2}(z+2i)^2} \right]$$

$$= \lim_{z \to 2i} \left[\frac{d}{dz} (z+2i)^{-2} \right]$$

$$= \lim_{z \to 2i} \left[-2(z+2i)^{-2-1} \right] (1)$$

$$= \lim_{z \to 2i} \left[\frac{-2}{(z+2i)^3} \right]$$

16×4
64

$$= \frac{-2}{(2i+2i)^3} = \frac{-2}{(4i)^3}$$

$$= \frac{+2}{+64i} = \frac{1}{32i}$$

$$\therefore \int_C \frac{dz}{(z^2+4)^2} = 2\pi i \, [\text{sum of residue}]$$

$$= 2\pi i \, [R_1 + R_2] = \cancel{2}\pi \cancel{i} \left[0 + \frac{1}{\cancel{32}\cancel{i}} \right]$$

16

$$= \pi/16$$

01/12/2022

Contour integration :

Type 1 : (integration around unit circle)

Note : $\int_0^{2\pi} \cos\theta, \sin\theta, d\theta$

$$\cos\theta = \frac{z^2+1}{2z}$$

$$\sin\theta = \frac{z^2-1}{2iz}$$

$$d\theta = \frac{dz}{iz}$$

1. Evaluate using contour integration for $\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$

Soln:

W.K.T

$$d\theta = \frac{dz}{iz}$$

$$\cos\theta = \frac{z^2+1}{2z} \quad \& \ |z|=1$$

$$\int_c \frac{\frac{dz}{iz}}{2+\left(\frac{z^2+1}{2z}\right)}$$

$$\int_c \frac{\frac{dz}{iz}}{\frac{4z+z^2+1}{2z}}$$

$$\int_c \frac{dz}{i\cancel{z}} \times \frac{2\cancel{z}}{z^2+4z+1}$$

$$\frac{2}{i}\int_c \frac{dz}{z^2+4z+1}$$

$$\therefore f(z) = \frac{1}{z^2+4z+1}$$

To find singular point

Given functions fails to be analytic

$\therefore$ denominator equal to zero

~~$|z|=1$~~

$z^2+4z+1=0$

$a=1,\ b=4,\ c=1$

$$f(z)=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$=\frac{-4\pm\sqrt{16-4}}{2(1)}$$

$$=\frac{-4\pm\sqrt{12}}{2}$$

$$=\frac{-4\pm\sqrt{4\times3}}{2}$$

$$=\frac{-4\pm2\sqrt{3}}{2}=\frac{-\cancel{4}^{2}}{\cancel{2}}\pm\frac{\cancel{2}\sqrt{3}}{\cancel{2}}$$

$\therefore z=-2\pm\sqrt{3}$

$z=-2+\sqrt{3},\ -2-\sqrt{3}$

$$f(z)=\frac{1}{(z+2+\sqrt{3})(z+2-\sqrt{3})}$$

Gn:

$|z|=1$

$$\begin{array}{r}2.00\\1.73\\\hline\end{array}$$

put $z=-2+\sqrt{3}$

$|-2+\sqrt{3}|=1$

$|-2+1.73|=1$

$|-0.27|=1$

$+0.27 < 1 \in C$

put $z = -2-\sqrt{3}$

$|-2-1.73| = 1$

$|-3.73| = 1$

$3.73 > 1 \notin C$

To find residues:

R1 $\quad z = -2-\sqrt{3} \notin C$

$\therefore \boxed{R_1 = 0}$

R2 $\quad z = -2+\sqrt{3}$ is a pole of order "1"

$$\left[\operatorname{Res} f(z)\right]_{z=-2+\sqrt{3}} = \lim_{z \to -2+\sqrt{3}} \cancel{(z+2-\sqrt{3})} \times \frac{1}{(z+2+\sqrt{3})\cancel{(z+2-\sqrt{3})}}$$

$$= \frac{1}{-\cancel{2}+\sqrt{3}+\cancel{2}+\sqrt{3}}$$

$$\boxed{R_2 = \frac{1}{2\sqrt{3}}}$$

$$\int_0^{2\pi} \frac{d\theta}{2+\cos\theta} = \frac{2}{i}\left[2\pi i\left[\text{Sum of residue}\right]\right]$$

$$= \frac{2}{i}\left[2\pi i\left(0+\frac{1}{2\sqrt{3}}\right)\right]$$

$$= \frac{\overset{2}{\cancel{4}}\pi \cancel{i}}{\cancel{2}\sqrt{3}\cancel{i}}$$

$$= \frac{2\pi}{\sqrt{3}}$$

Cauchy residues :-

1) Find the poles and residues of $\dfrac{z}{z^2-3z+2}$

Let $f(z) = \dfrac{z}{z^2-3z+2}$

$z^2 - 3z + 2 = 0$

$(z-1)(z-2) = 0$

$z = 1, 2$ are the simple poles

$f(z) = \dfrac{z}{(z-1)(z-2)}$

To find Residue:

i) $z=1$ of order 1

$$[\text{Res } f(z)]_{z=1} = \lim_{z \to 1} \cancel{(z-1)} \frac{z}{\cancel{(z-1)}(z-2)}$$

$$= \frac{1}{1-2}$$

$$= 1/-1 = -1$$

ii) $z=2$ of order 1

$$[\text{Res } f(z)]_{z=2} = \lim_{z \to 2} \cancel{(z-2)} \frac{z}{(z-1)\cancel{(z-2)}}$$

$$= \frac{2}{2-1} = \frac{2}{1} = 2$$

2) Evaluate $\int_C \frac{z-2}{z(z-1)} dz$ where C is the circle $|z|=2$.

Soln:

To find singular point:

Given function fails to be analytic

$\therefore$ denominator equal to zero

$\boxed{z=0}$ of order 1

$z-1=0$

$\boxed{z=1}$ of order 1

Gn:

$|z|=2$

$\boxed{z=0}$ $|0|=2$

$0<2 \in c$ (inside)

$\boxed{z=1}$ $|z|=2$

$|1|=2$

$1<2 \in c$ (inside)

To find residue:

R1 $z=0$

$$[\text{Res } f(z)]_{z=0} = \lim_{z\to 0} (\cancel{z}) \times \frac{(z-2)}{\cancel{z}(z-1)}$$

$$= \frac{0-2}{0-1}$$

$$= \frac{\cancel{-}2}{\cancel{-}1}$$

$$= 2$$

R_2

$z = 1$

$$[\text{Res } f(z)]_{z=1} = \lim_{z\to 1} \cancel{(z-1)} \times \frac{z-2}{z\cancel{(z-1)}}$$

$$= \frac{1-2}{1}$$

$$= -1$$

$$\therefore \int_c \frac{z-2}{z(z-1)}\, dz = 2\pi i\,[\text{sum of residue}]$$

$$= 2\pi i\,[R_1 + R_2]$$

$$= 2\pi i\,[2-1]$$

$$= 2\pi i\,[1]$$

$$\therefore \int_c \frac{z-2}{z(z-1)}\, dz = 2\pi i$$

Contour problem :-

2) using contour integration solve $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\, d\theta$

Soln

R.P of $z^2 = \cos 2\theta$

$$d\theta = \frac{dz}{iz}$$

$$\cos\theta = \frac{z^2+1}{2z}$$

$$\int_c \frac{\text{R.P of } z^2 \left(\frac{dz}{iz}\right)}{5 + \cancel{4}^{2}\left(\frac{z^2+1}{\cancel{2}z}\right)}$$

$$R.P \int_C \frac{z^2 \left(\frac{dz}{iz}\right)}{\frac{5z+2z^2+2}{z}}$$

$$R.P \int_C dz \left(\frac{z}{i}\right) \times \frac{z}{2z^2+5z+2}$$

$$\frac{R.P}{i} \int_C \frac{z^2}{2z^2+5z+2} dz$$

$$\therefore f(z) = \oint_C \frac{z^2}{2z^2+5z+2}$$

To find singular point:

Given function fails to be analytic

$\therefore$ denominator equal to zero.

$$2z^2+5z+2=0$$

4

4 5

$$z = -\frac{1}{2}, -\frac{4}{2}$$

$$\therefore z = -\frac{1}{2}, -2$$

$$f(z) = \frac{z^2}{2\left[\left(z+\frac{1}{2}\right)(z+2)\right]}$$

take co-eff for z^2 & take it into outside

W.K.T

$$|z| = 1$$

Put $z = -\frac{1}{2}$

$$\left|-\frac{1}{2}\right| = 1$$

$$0.5 < 1 \in C$$

Put $z = -2$

$|-2| = 1$

$2 > 1 \notin C$

To find residue:

$\underline{\underline{R_1}}$ $z = -2 \notin C$

$\therefore R_1 = 0$

$\underline{R_2}$

$z = -\frac{1}{2}$ is a pole of order "1"

$$[\text{Res } f(z)]_{z=-\frac{1}{2}} = \lim_{z \to -\frac{1}{2}} \cancel{(z+\frac{1}{2})} \times \frac{z^2}{2[\cancel{(z+\frac{1}{2})}(z+2)]}$$

$$= \frac{\left(-\frac{1}{2}\right)^2}{2\left(-\frac{1}{2}+2\right)}$$

$(-0.5)(-0.5)$

0.25

$$= \frac{1/4}{\cancel{2}\left(\frac{3}{\cancel{2}}\right)}$$

$$= \frac{1/4}{3} = \frac{1}{4} \times \frac{1}{3}$$

$$R_2 = \frac{1}{12}$$

$$\therefore \int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\, d\theta = \frac{R.P}{i}\left[2\pi i \,(\text{sum of Residue})\right]$$

$$= R.P\left[2\pi\left(0+\frac{1}{12}\right)\right]$$

$$= R.P\left[\frac{\pi}{6}\right]$$

$$= \frac{\pi}{6}$$

5) Evaluate $\int_0^{2\pi} \frac{d\theta}{13+5\sin\theta}$

Soln

$$d\theta = \frac{dz}{iz}$$

$$\sin\theta = \frac{z^2-1}{2iz}$$

$$\int_c \frac{\frac{dz}{iz}}{13+5\left(\frac{z^2-1}{2iz}\right)}$$

$$\int_c \frac{\frac{dz}{iz}}{\frac{26iz+5z^2-5}{2iz}}$$

$$\int_c \frac{dz}{\cancel{iz}} \times \frac{2\cancel{iz}}{26iz+5z^2-5}$$

$$2\int_c \frac{dz}{5z^2+26iz-5}$$

$$\therefore f(z) = \frac{1}{5z^2+26iz-5}$$

To find singular point:

Given function fails to be analytic

$\therefore$ denominator equal to zero

$$5z^2+26iz-5=0$$

$$a=5, \quad b=26i, \quad c=-5$$

$$f(z) = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{-26i \pm \sqrt{(26i)^2 - 4(5)(-5)}}{2(5)}$$

$$= \frac{-26i \pm \sqrt{-676+100}}{10}$$

$$= \frac{-26i \pm \sqrt{-576}}{10}$$

$$= \frac{-26i \pm 24i}{10}$$

$$\Rightarrow \frac{-26i+24i}{10}, \quad \frac{-26i-24i}{10}$$

$$\Rightarrow \frac{-2i}{10}, \quad \frac{-50i}{10}$$

$$\therefore z = -\frac{i}{5}, \; -5i$$

$$f(z) = \frac{1}{5\left[\left(z+\frac{i}{5}\right)(z+5i)\right]}$$

W.K.T

$$|z| = 1$$

Put $z = -\frac{i}{5}$

$$\left|-\frac{i}{5}\right| = 1$$

$$\sqrt{0^2 + \left(\frac{1}{5}\right)^2} = 1$$

$$\frac{1}{5} = 1$$

$$0.2 < 1 \in C$$

put $z = -5i$

$$|-5i| = 1$$

$$\sqrt{0^2+5^2} = 1$$

$$\sqrt{25} = 1$$

$$5 > 1 \notin C$$

To find residues :-

R_1 $\quad z = -5i \notin C$

$$\therefore R_1 = 0$$

R_2 $\quad z = -\frac{i}{5}$ of order 1

$$\left[Res\,(f(z))\right]_{z=-\frac{i}{5}} = \lim_{z \to -\frac{i}{5}} \cancel{\left(z+\frac{i}{5}\right)} \times \frac{1}{5\left[\cancel{\left(z+\frac{i}{5}\right)}(z+5i)\right]}$$

$$= \frac{1}{5\left[\frac{-i}{5}+5i\right]}$$

$$= \frac{1}{5\left[\frac{-i+25i}{5}\right]}$$

$$= \frac{1}{\cancel{5}\left[\frac{24i}{\cancel{5}}\right]}$$

$$R_2 = \frac{1}{24i}$$

$$\therefore \int_0^{2\pi} \frac{d\theta}{13+5\sin\theta} = 2\left[2\pi i\,(\text{sum of residues})\right]$$

$$= \cancel{4}\pi \cancel{i}\left(\frac{1}{\cancel{24}_{6}\cancel{i}}\right)$$

$$= \pi/6$$

4) Evaluate $\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta}$

Soln

$$d\theta = \frac{dz}{iz}$$

$$\sin\theta = \frac{z^2-1}{2iz}$$

$$\int_C \frac{(dz/iz)}{5-4\left(\frac{z^2-1}{2iz}\right)}$$

$$\int_C \frac{(dz/iz)}{\frac{5iz-2z^2+2}{iz}}$$

$$\int_C \frac{dz}{\cancel{iz}} \times \frac{\cancel{iz}}{5iz-2z^2+2}$$

$$\int_C \frac{dz}{-(2z^2-5iz-2)}$$

$$-1\int_C \frac{dz}{2z^2-5iz-2}$$

$$\therefore f(z) = \frac{dz}{2z^2-5iz-2}$$

To find singular point:

Given function fails to be analytic

$\therefore$ denominator equal to zero

$$2z^2-5iz-2=0$$

$a=2$, $b=-5i$, $c=-2$

$$z = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{5i \pm \sqrt{(-5i)^2 - 4(2)(-2)}}{2(2)}$$

$$= \frac{5i \pm \sqrt{-25+16}}{4}$$

$$= \frac{5i \pm \sqrt{-9}}{4}$$

$$= \frac{5i \pm 3i}{4}$$

$$z = \frac{5i+3i}{4}, \frac{5i-3i}{4}$$

$$= \frac{\overset{2}{\cancel{8}}i}{\cancel{4}}, \frac{\cancel{2}i}{\underset{2}{\cancel{4}}}$$

$$z = 2i, \frac{i}{2}$$

$$\therefore f(z) = \frac{1}{2\left[(z-2i)(z-\frac{i}{2})\right]}$$

W.K.T

$$|z| = 1$$

put $z = 2i$

$$|2i| = 1$$

$$\sqrt{0^2+(2)^2} = 1$$

$$\sqrt{4} = 1$$

$$2 > 1 \notin C$$

Put $z = \frac{i}{2}$

$$\left|\frac{i}{2}\right| = 1$$

$$\sqrt{0^2 + \left(\frac{1}{2}\right)^2} = 1$$

$$\sqrt{\frac{1}{4}} = 1$$

$$\frac{1}{2} = 1$$

$$0.5 < 1 \in C$$

To find residues:

<u>R1</u>

$$z = \cancel{\frac{i}{2}} 2i \notin C$$

$$\therefore R_1 = 0$$

<u>R2</u>

$z = \frac{i}{2}$ of order 1

$$\left[\text{Res } f(z)\right]_{z=\frac{i}{2}} = \lim_{z \to \frac{i}{2}} \cancel{\left(z - \frac{i}{2}\right)} \times \frac{1}{2\left[(z-2i)\cancel{\left(z - \frac{i}{2}\right)}\right]}$$

$$= \frac{1}{2\left[\frac{i}{2} - 2i\right]}$$

$$= \frac{1}{\cancel{2}\left[\frac{-3i}{\cancel{2}}\right]}$$

$$R_2 = -\frac{1}{3i}$$

$$\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta} = -\left[2\pi i\,(\text{sum of residues})\right]$$

$$= -\left[2\pi \cancel{i} \left(-\frac{1}{3\cancel{i}}\right)\right]$$

$$= \frac{2\pi}{3} \,//$$

5) Using contour integration solve $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta}$, $a>b>0$

Soln:

$$d\theta = \frac{dz}{iz}$$

$$\cos\theta = \frac{z^2+1}{2z}$$

$$\int_c \frac{(dz/iz)}{a+b\left(\frac{z^2+1}{2z}\right)}$$

$$\int_c \frac{(dz/iz)}{\frac{2az+bz^2+b}{2z}}$$

$$\int_c \frac{dz}{i\cancel{z}} \times \frac{2\cancel{z}}{2az+bz^2+b}$$

$$\frac{2}{i}\int_c \frac{dz}{bz^2+2az+b}$$

$$\therefore f(z) = \frac{1}{bz^2+2az+b}$$

To find singular point:

Given function fails to be analytic

∴ denominator equal to zero

$$bz^2 + 2az + b = 0$$

$$z = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$A = b$, $B = 2a$, $C = b$

$$z = \frac{-2a \pm \sqrt{4a^2 - 4b^2}}{2b}$$

$$= \frac{-2a \pm 2\sqrt{a^2 - b^2}}{2b}$$

$$z = \frac{-a + \sqrt{a^2 + b^2}}{b} , \frac{-a - \sqrt{a^2 - b^2}}{b}$$